Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 08

**** Unsupervised K-Means Clustering & Cluster Validation ****

Marks: /10

Date of Performance: / /2026 **Sign with Date:**

Aim: To implement the K-Means Clustering algorithm on an open-access multivariate dataset.

Lab Objective: To implement the K-Means Clustering algorithm on an open-access multivariate dataset, evaluate cluster cohesion and separation, and determine the optimal number of clusters (k∗) using both the Elbow Method (Within-Cluster Sum of Squares - WCSS) and the Silhouette Analysis Method.

Problem Statement:

Apply the K-Means clustering algorithm on a suitable open-access dataset to obtain the optimal number of clusters using the Elbow or Silhouette method.

1. Theory:

K-Means Objective Function (Inertia / WCSS)

K-Means partitions *n* observations into *k* distinct, non-overlapping clusters $C = \{C_1, C_2, \ldots, C_k\}$. It minimizes the Within-Cluster Sum of Squares (Inertia):

$$J(C) = \sum_{j-1}^{k} \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$$

Where μ_j is the centroid (mean coordinates) of cluster C_j.

2. Finding the Optimal *k*

1. The Elbow Method

- Evaluates total WCSS against a range of candidate cluster values (e.g., $k \in [2,10]$).
- As k increases, WCSS naturally decreases toward 0 (when $k=n$, each point is its own centroid).
- The **Elbow Point** represents the point of diminishing returns where adding another cluster yields only marginal reduction in inertia.

2. The Silhouette Analysis Method

Measures how similar a data point is to its own cluster compared to neighboring clusters. For sample i:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where:

- $a(i)$: Mean intra-cluster distance between sample i and all other points in the same cluster.
- $b(i)$: Mean nearest-cluster distance between sample i and the points in the closest neighboring cluster.

Interpretation:

- $s(i) \approx +1$: Point is far away from neighboring clusters (well clustered).
- $s(i) \approx 0$: Point lies on or close to the decision boundary between two clusters.
- $s(i) < 0$: Point is likely assigned to the wrong cluster.
- The optimal k maximizes the average Silhouette Score across all samples.

3. Dataset Selection: Mall Customer Segmentation

- **Source:** UCI ML Repository / Kaggle Open Access.
- **Samples (*N*):** 200 customer records.
- **Primary Clustering Features:**
    - Annual Income (k$)
    - Spending Score (1-100)
- **Physical Reason for Selection:** Provides distinct customer spending personas without synthetic overlap, making cluster validation visual and intuitive.

4. Algorithm & Execution Logic

INPUT : Unlabeled dataset X, Candidate cluster range K_range = [2, 10]
OUTPUT : Inertia values, Silhouette scores, Optimal k*, Final Cluster Labels

STEP 1: Load dataset and extract feature columns.
STEP 2: Apply StandardScaler to normalize dimensions so variance differences do not distort Euclidean distances.
STEP 3: For each k in K_range:
    3.1 Initialize KMeans(n_clusters=k, init='k-means++', random_state=42).
    3.2 Fit the model on standardized features.
    3.3 Record inertia_ (WCSS).
    3.4 Compute silhouette_score(X_scaled, labels).
STEP 4: Plot the Elbow Curve (k vs. Inertia) and identify the kink/elbow point.
STEP 5: Plot the Silhouette Score Curve (k vs. Average Silhouette Score) and locate the peak.
STEP 6: Train the final model using k* and visualize the 2D cluster partitions with their centroids.

Questions

1. What does the initialization parameter init='k-means++' do, and why is it preferred over random initialization?
2. Can the Elbow Method sometimes fail or give ambiguous results?
3. What does a negative Silhouette score for an individual data point indicate?
4. Why must features be standardized before applying K-Means?
5. What are the key limitations of the standard K-Means clustering algorithm?

Attendance and Timely Submission (10)	Execution (10)	Observations and Records (10)	Journal Documentation (10)	Oral (10)

Signature of Subject Teacher